Question	The process of excreting ammonia is called
Type	multiple choice
Option	Ammonotelic
Option	Ammonotelism
Option	Ureotelism
Option	Urecotelism
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Ammonia produced by metabolism is converted into urea in the
Type	multiple choice
Option	Kidney
Option	Liver
Option	Lungs
Option	Blood
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Nephridia are the tubular excretory structures of
Type	multiple choice
Option	Earthworms and other vertebrates
Option	Earthworms and other sponges
Option	Earthworms and other chordates
Option	Earthworms and other annelids
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Glomerulus along with Bowman's capsule is known as
Type	multiple choice
Option	Renal corpuscle
Option	Malpighian body
Option	Renal tubule
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following structures help to remove nitrogenous waste and in osmoregulation?
Type	multiple choice
Option	Nephridia
Option	Malpighian tubules
Option	Kidney
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Urea-excreting animals are commonly known as
Type	multiple choice

Option	Ammonotelic
Option	Ureotelic
Option	Ureotelism
Option	Both (b) and (c)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Minimum amount of water is required in the excretion of
Type	multiple choice
Option	Urea
Option	Uric acid
Option	Ammonia
Option	Ammonia and urea require same amount of water
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Kidney is used as an excretory organ in
Type	multiple choice
Option	Vertebrates
Option	Annelids
Option	Platyhelminthes
Option	Amphioxus
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Among the following nitrogenous wastes, the most toxic is
Type	multiple choice
Option	Ammonia
Option	Urea
Option	Uric acid
Option	CO_2
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Select the correct statement for Henle's loop
Type	multiple choice
Option	Connects PCT with glomerulus
Option	Starts after PCT and ends before DCT
Option	Absent in majority nephrons
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The Malpighian corpuscle, PCT and DCT are situated in
Type	multiple choice
Option	Medullary region
Option	Cortical region
Option	Malpighian body in cortex, PCT & DCT in medulla

Option	Malpighian body in medulla, PCT & DCT in cortex
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Select the correct statement about cortical nephrons
Type	multiple choice
Option	Loop of Henle is long
Option	Loop runs deep into medulla
Option	More in number
Option	Vasa recta present
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The efferent arteriole emerging from glomerulus forms
Type	multiple choice
Option	Vasa recta
Option	Peritubular capillaries
Option	Loop of Henle
Option	Renal pelvis
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Diameter of afferent arteriole is
Type	multiple choice
Option	Less than efferent arteriole
Option	More than efferent arteriole
Option	Equal to efferent arteriole
Option	Depends on blood flow
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Loop of Henle consists of
Type	multiple choice
Option	Ascending limb
Option	Descending limb
Option	Vasa recta
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**Urine formation involves** (i) Glomerular filtration (ii) Reabsorption (iii) Secretion (iv) Diffusion
Type	multiple choice
Option	i, ii, iv
Option	ii, iii, iv

Option	i, ii, iii
Option	i, ii, iii, iv
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	**Select the incorrect statement**
Type	multiple choice
Option	The first step in urine formation is the filtration of blood
Option	Filtration of blood is carried out by the glomerulus
Option	On an average, 1100–1200 ml of blood is filtered by the kidneys per minute
Option	100 percent
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**Filtration membrane consists of**
Type	multiple choice
Option	The epithelium of Bowman's capsule
Option	The endothelium of glomerular blood vessels
Option	A basement membrane between these two layers
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called**
Type	multiple choice
Option	Filtration pore
Option	Slit pore
Option	Pores of Kohn
Option	Both (a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**Kidney filters approximately**
Type	multiple choice
Option	1/5th part of cardiac output
Option	1/8th part of cardiac output
Option	1/10th part of cardiac output
Option	1/4th part of cardiac output
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**GFR in a healthy individual is approximately**
Type	multiple choice
Option	125 ml/minute
Option	180 ml/minute
Option	Both (a) and (b)
Option	250 ml/minute

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**About what percent of the filtrate has to be reabsorbed by the renal tubules?**
Type	multiple choice
Option	75 percent
Option	50 percent
Option	99 percent
Option	100 percent
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	**The osmolarity of medulla at the base of nephron is**
Type	multiple choice
Option	1200 mOsmol L^{-1}
Option	300 mOsmol L^{-1}
Option	600 mOsmol L^{-1}
Option	900 mOsmol L^{-1}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Osmolarity of cortex is**
Type	multiple choice
Option	Same as medulla
Option	Higher than medulla
Option	Less than medulla
Option	Same as urine
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	**Different osmolarity of cortex and medulla is maintained by**
Type	multiple choice
Option	NaCl and urea
Option	NaCl and ammonia
Option	Urea and HCl
Option	Ammonia only
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Human kidneys can produce urine nearly _____ than the initial filtrate formed**
Type	multiple choice
Option	Four times diluted
Option	Four times concentrated
Option	Two times concentrated
Option	Two times diluted
Answer	2

Solution	
Positive Marks	4
Negative Marks	1
Question	**In counter-current mechanism, the flow of filtrate in the two limbs of Henle's loop is in**
Type	multiple choice
Option	Opposite directions
Option	Same direction
Option	Sometimes in opposite direction
Option	Parallel direction
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Which segment of nephron allows passage of small amounts of urea into the medullary interstitium to keep up the osmolarity?**
Type	multiple choice
Option	PCT
Option	Collecting duct
Option	Bowman's capsule
Option	Glomerulus
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**The osmolarity of filtrate in PCT is**
Type	multiple choice
Option	1200 mOsmol L^{-1}
Option	300 mOsmol L^{-1}
Option	600 mOsmol L^{-1}
Option	900 mOsmol L^{-1}
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**Counter-current pattern is found between**
Type	multiple choice
Option	PCT and DCT
Option	DCT and loop of Henle
Option	DCT and vasa recta
Option	Loop of Henle and vasa recta
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**PCT is lined by**
Type	multiple choice
Option	Columnar epithelium
Option	Columnar ciliated epithelium
Option	Simple squamous epithelium
Option	Simple cuboidal brush border epithelium
Answer	4
Solution	

Positive Marks	4
Negative Marks	1
Question	**What percent of electrolytes is absorbed in PCT?**
Type	multiple choice
Option	70–80 percent
Option	10–20 percent
Option	50 percent
Option	2–5 percent
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Reabsorption is minimum in which part of nephron?**
Type	multiple choice
Option	PCT
Option	Henle's loop
Option	DCT
Option	Reabsorption is same in all parts of nephron
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**The ascending limb allows transport of electrolytes**
Type	multiple choice
Option	Actively
Option	Passively
Option	Both actively and passively
Option	Do not allow transport
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	**Site of conditional reabsorption is**
Type	multiple choice
Option	PCT
Option	DCT
Option	Loop of Henle
Option	All of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**DCT is capable of**
Type	multiple choice
Option	Reabsorption of HCO_3^-
Option	Selective secretion of hydrogen ions
Option	Selective secretion of potassium ions
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	**Select the correct statement about function of PCT**

Type	multiple choice
Option	Maintain the pH and ionic balance of the body fluids
Option	Selective secretion of hydrogen ions into blood
Option	Selective secretion of HCO_3^- into the filtrate
Option	All of the above
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Osmolarity of urine is**
Type	multiple choice
Option	1200 mOsmol L^{-1}
Option	300 mOsmol L^{-1}
Option	600 mOsmol L^{-1}
Option	900 mOsmol L^{-1}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Osmolarity of medulla is**
Type	multiple choice
Option	100 – 1200
Option	200 – 800
Option	600 – 1200
Option	50 – 1250
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	**Select the correct statement related to angiotensin II**
Type	multiple choice
Option	Increases GFR
Option	Decreases GFR
Option	Released from kidney
Option	Released from atria of heart
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**An increase in blood flow to the atria of the heart can cause the release of**
Type	multiple choice
Option	Angiotensin
Option	ANF
Option	Renin
Option	Both (a) and (c)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**ANF acts as**
Type	multiple choice
Option	Vasodilator
Option	Vasoconstrictor

Option	Increase blood pressure
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Blood pressure is decreased by the action of**
Type	multiple choice
Option	ADH
Option	ANF
Option	Aldosterone
Option	Angiotensin II
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	**JG cells release**
Type	multiple choice
Option	Renin
Option	ANF
Option	ADH
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	**Angiotensinogen is converted to Angiotensin I by**
Type	multiple choice
Option	Converting enzyme released from lungs
Option	ANF from heart
Option	Renin from kidney
Option	None of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1